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FAQs
MTH202

Question: What does it mean by the preservation of edge end point function in the definition of isomorphism of graphs?

Answer: Since you know that we are looking for two functions (Suppose one function is "f" and other function is "g") which preserve the edge end point function and this preservation means that if we have vi as an end point of the edge ej then f(vi) must be an end point of the edge g(ej) and also the converse that is if f(vi) be an end point of the edge g(ej) then we must have vi as an end point of the edge ej. Note that vi and ej are the vertex and edge of one graph respectively where as f (vi) and g (ej) are the vertex and edge in the other graph respectively.

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Question: Is there any method of identifying that the given graphs are isomorphic or not?(With out finding out two functions).

Answer: Unfortunately there is no such method which will identify whether the given graphs are isomorphic or not. In order to find out whether the two given graphs are isomorphic first we have to find out all the bijective mappings from the set vertices of one graph to the set of vertices of the other graph then find out all the bijective functions from the set of edges of one graph to the set of edges of the other graph. Then see which mappings preserve the edge end point function as defined in the definition of Isomorphism of graphs. But it is easy to identify that the two graphs are not isomorphic. First of all note that if there is any Isomorphic Invariant not satisfied by both the graphs, then we will say that the graphs are not Isomorphic. Note that if all the isomorphic Invariants are satisfied by two graphs then we can't conclude that the graphs are isomorphic. In order to prove that the graphs are isomorphic we have to find out two functions which satisfied the condition as defined in the definition of Isomorphism of graphs.

Question: What are Complementary Graphs?

Answer: Complementary Graph of a simple graph(G) is denoted by the (G bar) and has as many vertices as G but two vertices are adjacent in complementary

Graph by an edge if and only if these two vertices are not adjacent in G .

Question: What is the application of isomorphism in real word?

Answer: There are many applications of the graph theory in computer Science as well as in the Practical life; some of them are given below. (1) Now you also go through the puzzles like that we have to go through these points without lifting the pencil and without repeating our path. These puzzles can be solved by the Euler and Hamiltonian circuits. (2) Graph theory as well as Trees has applications in “DATA STRUCTURE" in which you will use trees, especially binary trees in manipulating the data in your programs. Also there is a common application of the trees is "FAMILY TREE”. In which we represent a family using the trees. (3) Another example of the directed Graph is "The World Wide Web ". The files are the vertices. A link from one file to another is a directed edge (or arc). These are the few examples.

Question: Are Isomorphic graphs are reflexive, symmetric and transitive?

Answer: We always talk about " RELXIVITY"" SYMMETRIC" and TRANSIVITY of a relation. We never say that a graph is reflexive, symmetric or transitive. But also remember that we draw the graph of a relation which is reflexive and symmetric and the property of reflexivity and symmetric is evident from the graphs, we can’t draw the graph of a relation such that transitive property of the relation is evident. Now consider the set of all graphs say it G, this being a set ,so we can define a relation from the set G to itself. So we define the relation of Isomorphism on the set G x G.(By the definition of isomorphism) Our claim is that this relation is an " Equivalence Relation" which means that the relation of Isomorphism’s of two graphs is "REFLEXIVE" "SYMMETRIC" and "TRANSITIVE". Now if you want to draw the graph of this relation, then the vertices of this graph are the graphs from the set G.

Question: Why we can't use the same color in connected portions of planar graph?

Answer: We define the coloring of graph in such a manner that we can’t assign the same color to the adjacent vertices because if we give the same colors to the adjacent vertices then they are indistinguishable. Also note that we can give the same color to the adjacent vertices but such a coloring is called improper coloring and the way which we define the coloring is known as the proper coloring. We are interested in proper coloring that’s why all the books consider the proper coloring

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Question: What is meant by isomorphic invariant?

Answer: A property "P" of a graph is known as Isomorphic invariant. if the same property is found in all the graphs which are isomorphic to it. And all these

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properties are called isomorphic invariant (Also it clear from the words Isomorphic Invariant that the properties which remain invariant if the two graphs are isomorphic to each other).

Question: What is an infinite Face?

Answer: When you draw a Planar Graph on a plane it divides the plane into different regions, these regions are known as the faces and the face which is not bounded by the edges of the graph is known as the Infinite face. In other words the region which is unbounded is known as Infinite Face.

Question: What is "Bipartite Graph"?

Answer: A graph is said to be Bipartite if it's set of vertices can be divided into two disjoint sets such that no two vertices of the same set are adjacent by some edge of the graph. It means that the edges of one set will be adjacent with the vertices of the other set.

Question: What is chromatic number?

Answer: While coloring a graph you can color a vertex which is not adjacent with the vertices you already colored by choosing a new color for it or by the same color which you have used for the vertices which are not adjacent with this vertex. It means that while coloring a graph you may have different number of colors used for this purpose. But the least number of colors which are being used during the coloring of Graphs is known as the Chromatic number.

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Question: What is the role of Discrete mathematics in our prectical life. what advantages will we get by learning it.

Answer: In many areas people have to faces many mathematical problems which can,t be solved in computer so discrete mathematics provide the facility to overcome these problems. Discrete math also covers the wide range of topics, starting with the foundations of Logic, Sets and Functions. It moves onto integer mathematics and matrices, number theory, mathematical reasoning, probability graphs, tree data structures and Boolean algebra.So that is why we need discrete math.

Question: What is the De Morgan's law .

Answer: De Morgan law states " Negation of the conjunction of two statements is logiacally equivalent to the disjunction of their negation and Negation of the disjunction of two statements is logically equivalent to the conjucnction of their negation". i.e. ~(p^q) = ~p v ~q and ~(p v q)= ~p ^ ~q For example: " The bus was late and jim is waiting "(this is an example of conjuction of two statements) Now apply neaggation on this statement you will get through De

Morgan's law " The bus was not late or jim is not waiting" (this is the disjunction of negation of two statements). Now see both statement are logically equivalent.Thats what De Morgan want to say

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Question: What is Tauology?

Answer: A tautology is a statement form that is always true regardless of the truth values of the statement variables. i.e. If you want to prove that (p v q) is tautology ,you have to show that all values of statement (p v q) are true regardless of the values of p and q.If all the values of the satement (p v q) is not true then this statement is not tautology.

Question: What is binary relations and reflexive,symmetric and transitive.

Answer: Dear student! First of all ,I will tell you about the basic meaning of relation i.e It is a logical or natural association between two or more things; relevance of one to another; the relation between smoking and heart disease. The connection of people by blood or marriage. A person connected to another by blood or marriage; a relative. Or the way in which one person or thing is connected with another: the relation of parent to child. Now we turn to its mathematically definition, let A and B be any two sets. Then their cartesian product (or the product set) means a new set "A x B " which contains all the ordered pairs of the form (a,b) where a is in set A and b is in set B. Then if we take any subset say 'R' of "A x B" ,then 'R' is called the binary relation. Note All the subsets of the Cartesian product of two sets A and B are called the binary relations or simply a relation,and denoted by R. And note it that one raltion is also be the same as "A x B". Example: Let A={1,2,3} B={a,b} be any two sets. Then their Cartesian product means "A x B"={ (1,a),(1,b),(2,a),(2,b),(3,a),(3,b) } Then take any set which contains in "A x B" and denote it by 'R'. Let we take R={(2,b),(3,a),(3,b)} form "A x B". Clearly R is a subset of "A x B" so 'R' is called the binary relation. A reflexive relation defined on a set say ‘A’ means “all the ordered pairs in which 1st element is mapped or related to itself.” For example take a relation say R1= {(1,1), (1,2), (1,3), (2,2) ,(2,1), (3,1) (3,3)} from “A x B” defined on the set A={1,2,3}. Clearly R1 is reflexive because 1,2 and 3 are related to itself. A relation say R on a set A is symmetric if whenever aRb then bRa,that is ,if whenever (a,b) belongs to R then (b,a) belongs to R for all a,b belongs to A. For example given a relation which is R1={(1,1), (1,2), (1,3), (2,2) ,(2,1), (3,1) (3,3)} as defined on a set A={1,2,3} And a relation say R1 is symmetric if for every (a, b) belongs to R ,(b, a) also belongs to R. Here as (a, b)=(1,1) belongs to R then (b, a)=(1,1)also belongs to R. as (a,b)=(1,2) belongs to R then (b,a)=(2,1)also belongs to R. as (a,b)=(1,3) belongs to R then (b,a)=(3,1)also belongs to R.etc So clearly the above relation R is symmetric. And read the definition of transitive relation from the handouts and the book. You can easily understand it.

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Question: What is the matrix relation .

Answer: Suppose that A and B are finite sets.Then we take a relation say R from A to B. From a rectangular array whose rows are labeled by the elements of A and whose columns are labeled by the elements of B. Put a 1 or 0 in each position of the array according as a belongs to A is or is not related to b belongs to B. This array is called the matrix of the relation. There are matrix relations of reflexive and symmetric relations. In reflexive relation, all the diagonal elements of relation should be equal to 1. For example if R = {(1,1), (1,3), (2,2), (3,2), (3,3)} defined on A = {1,2,3}. Then clearly R is reflexive. Simply in making matrix relation In the above example,as the defined set is A={1,2,3} so there are total three elements. Now we take 1, 2 and 3 horizontally and vertically.i.e we make a matrix from the relation R ,in the matrix you have now 3 columns and 3 rows. Now start to make the matrix ,as you have first order pair (1, 1) it means that 1 maps on itself and you write 1 in 1st row and in first column. 2nd order pair is (1, 3) it means that arrow goes from 1 to 3.Then you have to write 1 in 1st row and in 3rd column. (2, 2) means that arrow goes from 2 and ends itself. Here you have to write 1 in 2nd row and in 2nd column. (3,2) means arrow goes from 3 and ends at 2. Here you have to write 1 in 3rd row and in 2nd column. (3, 3) means that 3 maps on itself and you write 1 in 3rd row and in 3rd column. And where there is space empty or unfilled ,you have to write 0 there.

Question: what is binary relation.

Answer: Let A and B be any two sets. Then their cartesian product(or the product set) means a new set "A x B " which contains all the ordered pairs of the form (a,b) where a is in set A and b is in set B. Let we take any subset say 'R' of "A x B" ,then 'R' is called the binary relation. Note it that 'R' also be the same as "A x B". For example: Let A={1,2,3} B={a,b} be any two sets. Then their cartesian product means "A x B"={ (1,a),(1,b),(2,a),(2,b),(3,a),(3,b) } Then take any set which contains in "A x B" and denote it by 'R'. Let R={(2,b),(3,a),(3,b)} Clearly R is a subset of "A x B" so 'R' is called the binary relation.

Question: Role of "Discrete Mathematics" in our prectical life. what advantages will we get by learning it.

Answer: Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. While the ideas of calculus were fundamental to the science and technology of the industrial revolution, the ideas of discrete mathematics underline the science and technology specific to the computer age. Logic and proof: An important goal of discrete mathematics is to develop students' ability to think abstractly. This requires that students learn to use logically valid forms of argument, to avoid common logical errors, to understand what it means to reason from

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definition, and to know how to use both direct and indirect argument to derive new results from those already known to be true. Induction and Recursion: An exciting development of recent years has been increased appreciation for the power and beauty of “recursive thinking”: using the assumption that a given problem has been solved for smaller cases, to solve it for a given case. Such thinking often leads to recurrence relations, which can be “solved” by various techniques, and to verifications of solutions by mathematical induction. Combinatorics: Combinatorics is the mathematics of counting and arranging objects. Skill in using combinatorial techniques is needed in almost every discipline where mathematics is applied, from economics to biology, to computer science, to chemistry, to business management. Algorithms and their analysis: The word algorithm was largely unknown three decades ago. Yet now it is one of the first words encountered in the study of computer science. To solve a problem on a computer, it is necessary to find an algorithm or step-by-step sequence of instructions for the computer to follow. Designing an algorithm requires an understanding of the mathematics underlying the problem to be solved. Determining whether or not an algorithm is correct requires a sophisticated use of mathematical induction. Calculating the amount of time or memory space the algorithm will need requires knowledge of combinatorics, recurrence relations functions, and O-notation. Discrete Structures: Discrete mathematical structures are made of finite or count ably infinite collections of objects that satisfy certain properties. Those are sets, bolean of algebras, functions, finite start automata, relations, graphs and trees. The concept of isomorphism is used to describe the state of affairs when two distinct structures are the same intheir essentials and diffr only in the labeling of the underlying objects. Applications and modeling: Mathematics topic are best understood when they are seen ina variety of contexts and used to solve problems in a broad range of applied situations. One of the profound lessons of mathematics is that the same mathematical model can be used to solve problems in situations that appear superficially to be totally dissimilar. So in the end i want to say that discrete mathematics has many uses not only in computer science but also in the other fields too.

Question: what is the basic difference b/w sequences and series

Answer: A sequence is just a list of elements .In sequnce we write the terms of sequence as a list (seperated by comma's). e.g 2,3,4,5,6,7,8,9,... (in this we have terms 2,3,4,5,6,7,8,9 and so on).we write these in form of list seperated by comma's. And the sum of the terms of a sequence forms a series. e.g we have sequence 1,2,3,4,5,6,7 Now the series is sum of terms of sequence as 1+2+3+4+5+6+7.

Question: what is the purpose of permutations?

Answer: Permutation is an arrangement of objects in a order where repitition is not allowed. We need arrangments of objects in real life and also in

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mathematical problems.We need to know in how many ways we can arrange certain objects. There are four types of arrangments we have in which one is permutation.

Question: what is inclusion-exclusion principle

Answer: Inclusion-Exclusion principle contain two rules which are If A and B are disjoint finite sets, then n(AÈB) = n(A) + n(B) And if A and B are finite sets, then n(AÈB) = n(A) + n(B) - n(AÇB) For example If there are 15 girls students and 25 boys students in a class then how many students are in total. Now see if we take A ={ 15 girl students} and B={ 25 boys students} Here A and B are two disjoints sets then we can apply first rule n(AÈB) = n(A) + n(B) =15 + 25 =40 So in total there are 40 students in class. Take another Example for second rule. How many integers from 1 through 1000 are multiples of 3 or multiples of 5. Let A and B denotes the set of integers from 1 through 1000 that are multiples of 3 and 5 respectivly. n(A)= 333 n(B)=200 But these two sets are not disjoint because in A and B we have those elements which are multiple of both 3 and 5. so n(AÇB) =66 n(AÈB) = n(A) + n(B) - n(AÇB) =333 + 200 - 66 = 467

Question: How to use conditional probobility

Answer: Dear student In Conditional probability we put some condition on an event to be occur. e.g. A pair of dice is tossed. Find the probability that one of the dice is 2 if the sum is 6. If we have to find the probability that one of the dice is 2, then it is the case of simple probability. Here we put a condition that sum is six. Now A = { 2 appears in atleast one die} E = {sum is 6 } Here E = { (1,5), (2, 4), (3, 3), (4, 2), (5, 1) } Here two order pairs (2, 4) and (4, 2) satisfies the A. (i.e. belongs to A) Now A (intersection) B= { (2,4), (4,2)} Now by formula P(A/E) = P(A (intersection) E)/ P(E) = 2/5

Question: In which condition we use combination and in which condition permutation.

Answer: This depends on the statement of question. If in the statement of question you finds out that repetition of objects are not allowed and order matters then we use Permutation. e.g. Find the number of ways that a party of seven persons can arrange themselves in a row of seven chairs. See in this question repetition is no allowed because whenever a person is chosen for a particular seat r then he cannot be chosen again and also order matters in the arrangements of chairs so we use permutation here. If in the question repetition of samples are not allowed and order does not matters then we use combination. A student is to answer eight out of ten questions on an exam. Find the number m of ways that the student can choose the eight questions See in this question repetition is not allowed that is when you choose one question then you cannot choose it again and also order does not matters(i.e either he solved Q1 first or Q2 first) so you use combination in this question.

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Question: What is the differnce between edge and vertex

Answer: Vertices are nodes or points and edges are lines/arcs which are used to connect the vertices. e.g If you are making the graph to find the shortest path or for nay purpose of cites and roads between them which contain Lahore, Islamabad, Faisalabad , Karachi, and Multan. Then cities Lahore, Islamabad, Faisalabad , Karachi, and Multan are vertices and roads between them are edges.

Question: What is the differnce between yes and allowed in graphs.

Answer: Allowed mean that specific property can be occurs in that case but yes mean that specific property always occurs in that case. e.g. In Walk you may start and end at same point and may not be (allowed). But in Closed Walk you have to start and end at same point (yes).

Question: what is the meanging of induction? and also Mathematical Induction?

Answer: Basic meaning of induction is: a)The act or an instance of inducting. b) A ceremony or formal act by which a person is inducted, as into office or military service. In Mathematics. A two-part method of proving a theorem involving a positive integral variable. First the theorem is verified for the smallest admissible value of the integer. Then it is proven that if the theorem is true for any value of the integer, it is true for the next greater value. The final proof contains the two parts. As you have studied. It also means that presentation of material, such as facts or evidence, in support of an argument or a proposition. Whether in Physics Induction means the creation of a voltage or current in a material by means of electric or magnetic fields, as in the secondary winding of a transformer when exposed to the changing magnetic field caused by an alternating current in the primary winding. In Biochemistry,it means that the process of initiating or increasing the production of an enzyme or other protein at the level of genetic transcription. In embryology,it means that the change in form or shape caused by the action of one tissue of an embryo on adjacent tissues or parts, as by the diffusion of hormones or chemicals.

Question: what is one to one function .

Answer: Dear student! First of all read the definition of function i.e. Function:It is a rule or a machine from a set X to a set Y in which each element of set X maps into the unique element of set Y. Let we take a function from set X to set Y like f: X----Y defined as f(x) =y. Here x is called the pre-image of y and y is called the image of x under the rule f. One-One function:means a function in which distinct elements of set X have their distinct images in set Y. i.e. any element of set X have not more than one image in set Y,if such

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happen then this is mapping is not one-one function. Like f: X –-Y from R to R such that f(x)= x ^3(i.e. x raise to power 3) This function is one –one because for f(1)=1 raise to power 3=1 (1 is the image of 1 under the rule f) f(2)=2 raise to power 3=8 (8 is the image of 2 under the rule f) f(3)=3 raise to power 3=27(27 is the image of 3 under the rule f) etc Clearly every element of X have distinct images in set Y. But f(x) = x^4 from R---R is not one-one, because 1 has the image 1,also -1 has the image 1. So as 1 and -1 has the same images that's the way it is called not one-one. There are certain conditions for checking any mapping to be one-one or not. Let f(x1)=f(x2) thenprove x1=x2. Or let x1 is not equal to x1 .then prove f(x1)is not equal to f(x2). In all these cases, such mapping (or functions)are called one-one functions

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Question: what is the meaning of implication .

Answer: Implication means "Condition of being implied". We put some condition on a statement to be true or false. e.g. If you got A in class then you will go for tour. We put a condition that for going on tour you should get A.

Question: What is converse,Inverse and contrapositive.

Answer: Inverse, converse and contrapositive are also statements which are made from simple conditional statement. The converse of the conditional statement p --> q is q --> p . i.e. now conclusion become hypothesis and hypothesis become conclusion. The contrapositive of the conditional statement p --> q is ~ q --> ~ p i.e. Take the negation of conclusion and made it hypothesis and similarly take the negation of conclusion and made it conclusion. The inverse of the conditional statement p --> q is ~p --> ~q

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Question: What is the diffirence between "Valid Argument" and "Modus Panes"?

Answer: An argument is an assertion that the given set of propositions called premises yields another proposition called conclusion. Now an argument is said to be valid if conclusion is true whenever all the premises are true. Modus ponens is a law which states that if both an implication and its hypothesis are true then the conclusion of this implication is true. This defines a law that if in an argument first premises is implication and second is hypothesis and both are true then the conclusion is always true. This is a specific case of a valid argument.

Question: What is the diffirence between "Modus Tollens" and "Modus Panes"?

Answer: Modus ponens is a law that if the argument is of the form p p---> q Therefore q Then it'll always be a tautology. i.e. if the hypothesis and and its implication are known to be true then the conclusion of this implication is

true. Similar is the case with modus tollens that is if the argument is the of the type ~ q P-->q Therefore ~ p Then it'll always tautology. Both are laws which have different set of premises and conclusion but both are tautology. So the difference is between their premises and conclusion.

Question: What is "Hypothetical Syllogism".

Answer: Hypothetical syllogism is a law that if the argument is of the form p --> q q--> r Therefore p---> r Then it'll always be a tautology. i.e. if the p implies q and q implies r is true then its conclusion p implies r is always true.

Question: A set is define a well define collection of distinct objects so why an empty set is called a set although it has no element?

Answer: Some time we have collection of zero objects and we call them empty sets. e.g. Set of natural numbers greater than 5 and less than 5. A = { x belongs to N / 5< x < 5 } Now see this is a set which have collection of elements which are greater than 5 and less than 5 (from natural number).

Question: What is improper subset.

Answer: Let A and B be sets. A is proper subset of B, if, and only if, every element of A is in B but there is at least on element if B that is not in A. Now A is improper subset of B, if and only if, every element of A is in B and there is no element in B which is not in A. e.g. A= { 1, 2 , 3, 4} B= { 2, 1, 4, 3} Now A is improper subset of B. Because every element of A is in B and there is no element in B which is not in A

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Question: How to check validity and unvalidity of argument through diagram.

Answer: To check an argument is valid or not you can also use Venn diagram. We identify some sets from the premises . Then represent those sets in the form of diagram. If diagram satisfies the conclusion then it is a valid argument otherwise invalid. e.g. If we have three premises S1: all my friends are musicians S2: John is my friend. S3: None of my neighbor are musicians. conclusion John is not my neighbor. Now we have three sets Friends, Musicians, neighbors. Now you see from premises 1 and 2 that friends are subset of musicians .From premises 3 see that neighbor is an individual set that is disjoint from set musicians. Now represent then in form of Venn diagram. Musicians neighbour Friends Now see that john lies in set friends which is disjoint from set neighbors. So their intersection is empty.Which shows that john is not his neighbor. In that way you can check the validity of arguments

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Question: why we used venn digram?

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common.

Question: what is composite relation .

Answer: Let A, B, and C be sets, and let R be relation from A to B and let S be a relation from B to C. Now by combining these two relations we can form a relation from A to C. Now let a belongs to A, b belongs to B, and c belongs to C. We can write relations R as a R b and S as b S c. Now by combining R and S we write a (R 0 S) c . This is called composition of Relations holding the condition that we must have a b belongs to B which can be write as a R b and b S c (as stated above) . e.g. Let A= {1,2,3,4}, B={a,b,c,d} , C ={x,y,z} and let R={ {1,a), (2, d), (3, a), (3, b), (3, d) } and S={ (b, x), (b, z), (c, y), (d, z)} Now apply that condition which is stated above (that in the composition R O S only those order pairs comes which have earlier an element is common in them e.g. from R we have (3, b) and from S we have (b, x) .Now one relation relate 3 to b and other relates b to x and our composite relation omits that common and relates directly 3 to x.) I do not understand your second question send it again. Now R O S ={(2,z), (3,x), (3,z)}

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Question: What are the conditions to confirm functions .

Answer: The first condition for a relation from set X to a set Y to be a function is 1.For every element x in X, there is an element y in Y such that (x, y) belongs to F. Which means that every element in X should relate with distinct element of Y. e.g if X={ 1,2,3} and Y={x, y} Now if R={(1,x),(2,y),(1,y),(2,x)} Then R will not be a function because 3 belongs to X but is does not relates with any element of Y. so R={(1,x),(2,y),(3,y)} can be called a function because every element of X is relates with elements of Y. Second condition is : For all elements x in X and y and z in Y, if (x, y) belongs to F and (x, z) belongs to F, then y = z Which means that every element in X only relates with distinct element of Y. i.e. R={(1,x),(2,y),(2,x), (3,y)} cannot be called as function because 2 relates with x and y also.

Question: When a function is onto.

Answer: First you have to know about the concept of function. Function:It is a rule or a machine from a set X to a set Y in which each element of set X maps into the unique element of set Y. Onto Function: Means a function in which every element of set Y is the image of at least one element in set X. Or there

should be no element left in set Y which is the image of no element in set X. If such case does not exist then the function is not called onto. For example:Let we define a function f : R----R such that f(x)=x^2 (where ^ shows the symbol of power i.e. x raise to power 2). Clearly every element in the second set is the image of atleast one element in the first set. As for x=1 then f(x)=1^2=1 (1 is the image of 1 under the rule f) for x=2 then f(x)=2^2=4 (4 is the image of 2 under the rule f) for x=0 then f(x)=0^2=0 (0 is the image of 0 under the rule f) for x=-1 then f(x)=(-1)^2=1 (1 is the image of -1 under the rule f) So it is onto function.

Question: Is Pie an irrational number?

Answer: Pi π is an irrational number as its exact value has an infinite decimal expansion: Its decimal expansion never ends and does not repeat.

The numerical value of π truncated to 50 decimal places is:

3.14159 26535 89793 23846 26433 83279 50288 41971 69399 37510

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Question: Difference between sentence and statement.

Answer: A sentence is a statement if it have a truth value otherwise this sentence is not a statement.By truth value i mean if i write a sentence "Lahore is capital of Punjab" Its truth value is "true".Because yes Lahore is a capital of Punjab. So the above sentence is a statement. Now if i write a sentence "How are you" Then you cannot answer in yes or no.So this sentence is not a statement. Every statement is a sentence but converse is not true.

Question: What is the truth table?

Answer: Truth table is a table which describe the truth values of a proposition. or we can say that Truth table display the complete behaviour of a proposition. There fore the purpose of truth table is to identify its truth values. A statement or a proposition in Discrete math can easily identify its truth value by the truth table. Truth tables are especially valuable in the determination of the truth values of propositions constructed from simpler propositions. The main steps while making a truth table are "first judge about the statement that how much symbols(or variables) it contain. If it has n symbols then total number of combinations=2 raise to power n. These all the combinations give the truth value of the statement from where we can judge that either the truthness of a statement or proposiotion is true or false. In all the combinations you have to put values either "F" or "T" against the variales.But note it that no row can be repeated. For example "Ali is happy and healthy" we denote "ali is happy" by p and "ali is healthy" by q so the

above statement contain two variables or symbols. The total no of combinations are =2 raise to power 2(as n=2) =4 which tell us the truthness of a statement.

Question: how empty set become a subset of every set.

Answer: If A & B are two sets, A is called a subset of B, if, and only if, every element of A is also an element of B. Now we prove that empty set is subset of any other set by a contra positive statement(of above statement) i.e. If there is any element in the the set A that is not in the set B then A is not a subset of B. Now if A={} and B={1,3,4,5} Then you cannot find an element which is in A but not in B. So A is subset of B.

Question: What is rational and irrational numbers.

Answer: A number that can be expressed as a fraction p/q where p and q are integers and q\not=0, is called a rational number with numerator p and denominator q. The numbers which cannot be expressed as rational are called irrational number. Irrational numbers have decimal expansions that neither terminate nor become periodic where in rational numbers the decimal expansion either terminate or become periodic after some numbers.

Question: what is the difference between graphs and spanning tree?

Answer: First of all, a graph is a "diagram that exhibits a relationship, often functional, between two sets of numbers as a set of points having coordinates determined by the relationship. Also called plot". Or A pictorial device, such as a pie chart or bar graph, used to illustrate quantitative relationships. Also called chart. And a tree is a connected graph that does not contain any nontrivial circuit. (i.e., it is circuit-free) Basically, a graph is a nonempty set of points called vertices and a set of line segments joining pairs of vertices called edges. Formally, a graph G consists of two finite sets: (i) A set V=V(G) of vertices (or points or nodes) (ii) A set E=E(G) of edges; where each edge corresponds to a pair of vertices. Whereas, a spanning tree for a graph G is a subgraph of G that contains every vertex of G and is a tree. It is not neccesary for a graph to always be a spanning tree. Graph becomes a spanning tree if it satisfies all the properties of a spanning tree.

Question: What is the probability ?

Answer: The definition of probability is : Let S be a finite sample space such that all the outcomes are equally likely to occur. The probability of an event E, which is a subset of S, is P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) This definition is due to ‘Laplace.’ Thus probability is a concept which measures numerically the degree of certainty or uncertainty of the occurrence of an event. Explaination The basic steps of probability that u have to remember are as under 1. First

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list out all possible out comes. That is called the sample space S For example when we roll a die the all possible outcomes are the set S i.e. S = {1,2,3,4,5,6} 2. Secondly we have to find out all that possible outcomes, in which the probability is required . For example we are asked to find the probability of even numbers. First we decide any name of that event i.e E Now we check all the even numbers in S which are E = {2,4,6} Remember Event is always a sub-set of Sample space S. 3. Now we apply the definition of probability P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) So from above two steps we have n (E) = 3 and n (S) = 6 then P(E) = 3 / 6 = 1/2 which is probability of an even number.

Question: what is permutation?

Answer: Permutation comes from the word permute which means " to change the order of." Basically permutation means a "complete change." Or the act of altering a given set of objects in a group. In Mathematics point of view it means that a ordered arrangement of the elements of a set (here the order of elements matters but repetition of the elements is not allowed).

Question: What is a function.

Answer: A function say 'f' is a rule or machine from a set A to the set B if for every element say a of A, there exist a unique element say b of set such that b=f(a) Where b is the image of a under f,and a is the pre-image. Note it that set A is called the domain of f and Y is called the codomain of f. As we know that function is a rule or machine in which we put an input,and we get an output.Like that a juicer machine.We take some apples(here apples are input) and we apply a rule or a function of juicer machine on it,then we get the output in the form of juice.

Question: What is p implies q.

Answer: p--- >q means to "go from hypothesis to a conclusion" where p is a hypothesis and q is a conclusion. And note it that this statement is conditioned because the "truth ness of statement p is conditioned on the truth ness of statement q". Now the truth value of p--->q is false only when p is true and q is false otherwise it will always true. E.g. consider an implication "if you do your work on Sunday ,I will give you ten rupees." Here p=you do your work on Sunday (is the hypothesis) , q=I will give you ten rupees (the conclusion or promise). Now the truth value of p---->q will false only when the promise is braked. i.e. You do your work on Sunday but you do not get ten rupees. In all other conditions the promise is not braked.

Question: What is valid and invalid arguments.

Answer: As "an argument is a list of statements called premises (or assumptions or

hypotheses) which is followed by a statement called the conclusion. " A valid argument is one in which the premises entail(or imply) the conclusion. 1)It cannot have true premises and a false conclusion. 2)If its premises are true, its conclusion must be true. 3)If its conclusion is false, it must have at least one false premise. 4)All of the information in the conclusion is also in the premises. And an invalid agrument is one in which the premises do not entail(or imply) the conclusion. It can have true premises and a false conclusion. Even if its premises are true, it may have a false conclusion. Even if its conclusion is false, it may have true premises. There is information in the conclusion that is not in the premises. To know them better,try to solve more and more examples and exercises.

Question: What is domain and co -domain.

Answer: Domain means "the set of all x-coordinates in a relation". It is very simple,Let we take a function say f from the set X to set Y. Then domain means a set which contain all the elements of the set X. And co domain means a set which contain all the elements of the set Y. For example: Let we define a function "f" from the set X={a,b,c,d} to Y={1,2,3,4}. such that f(a)=1, f(b)=2, f(c)=3, f(d)=1 Here the domain set is {a,b,c,d} And the co-domain set is {1,2,3,4} Where as the image set is {1,2,3}.Because f(a)=1 as 1 is the image of a under the rule 'f'. f(b)=2 as 2 is the image of b under the rule 'f'. f(c)=3 as 3 is the image of c under the rule 'f'. f(d)=1as 1 is the image of d under the rule 'f'. because "image set contains only those elements which are the images of elements found in set X". Note it that here f is one -one but not onto,because there is one element '4' left which is the image of nothing element under the rule 'f'.

Question: What is the difference between k-sample,k-selection, k-permutation and k-combination?

Answer: Actually, these all terms are related to the basic concept of choosing some elements from the given collection.

For it, two things are important:

1) Order of elements .i.e. which one is first, which one is second and so on.
2) Repetition of elements

So we can get 4 kinds of selections:

1) The elements have both order and repetition. (It is called k-sample)
2) The elements have only order, but no repetition. (It is called k-permutation)
3) The elements have only repetition, but no order. (It is called k-selection)

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4) The elements have no repetition and no order. (It is called k-combination)

Question: What is a combination?

Answer: A combination is an un-ordered collection of unique elements. Given S, the set of all possible unique elements, a combination is a subset of the elements of S. The order of the elements in a combination is not important (two lists with the same elements in different orders are considered to be the same combination). Also, the elements cannot be repeated in a combination (every element appears uniquely once

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Question: What is the main deffernce between Calculus and Discrete Maths?

Answer: Discrete mathematics is the study of mathematics which concerns to the study of discrete objects. Discrete math build students approach to think abstractly and how to handle mathematical models problems in computer While Calculus is a mathematical tool used to analyze changes in physical quantities. Or "Calculus is sometimes described as the mathematics of change." Also calculus played an important role in industrial area as well discrete math in computer.

Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. the ideas of discrete mathematics underline the science and technology specific to the computer age. An important goal of discrete mathematics is to develop students' ability to think abstractly.

Question: What is the use of kruskal's algorithn in our daily life?

Answer: The Kruskal's algorithm is usually used to find minimum spanning tree i.e. the possible smallest tree that contains all the vertices. The standard application is to a problem like phone network design. Suppose, you have a business with several offices; you want to lease phone lines to connect them up with each other; and the phone company charges different amounts of money to connect different pairs of cities. You want a set of lines that connects all your offices with a minimum total cost. It should be a spanning tree, since if a network isn't a tree you can always remove some edges and save money. A less obvious application is that the minimum spanning tree can be used to approximately solve the traveling salesman problem. A convenient formal way of defining this problem is to find the shortest path that visits each point at least once.

Question: What is irrational number?

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Answer: Irrational number An irrational number can not be expressed as a fraction. In decimal form, irrational numbers do not repeat in a pattern or terminate. They "go on forever" (infinity). Examples of irrational numbers are: pi= 3.141592654...

Question: Define membership table and truth table.

Answer: Membership table: A table displaying the membership of elements in sets. Set identities can also be proved using membership tables. An element is in a set, a 1 is used and an element is not in a set, a 0 is used. Truth table: A table displaying the truth values of propositions.

Question: What is sequence and series?

Answer: Sequence A sequence of numbers is a function defined on the set of positive integer. The numbers in the sequence are called terms. Another way, the sequence is a set of quantities u1, u2, u3... stated in a definite order and each term formed according to a fixed pattern. U r =f(r) In example: 1,3,5,7,... 2,4,6,8,... 1 2 ,− 2 2 ,3 2 ,− 4 2 ,... Infinite sequence:- This kind of sequence is unending sequence like all natural numbers: 1, 2, 3, ... Finite sequence:- This kind of sequence contains only a finite number of terms. One of good examples are the page numbers. Series:- The sum of a finite or infinite sequence of expressions. 1+3+5+7+...

Question: What is Euclidean algorithm?

Answer: In number theory, the **Euclidean algorithm** (also called **Euclid's algorithm**) is an algorithm to determine the greatest common divisor (GCD) of two integers.

Its major significance is that it does not require factoring the two integers, and it is also significant in that it is one of the oldest algorithms known, dating back to the ancient Greeks.

Question: what is the circle definition?

Answer: A circle is the locus of all points in a plane which are equidistant from a fixed point. The fixed point is called centre of that circle and the distance is called radius of that circle

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Question: What is meant by Discrete?

Answer:

A type of data is discrete if there are only a finite number of values possible. Discrete data usually occurs in a case where there are only a certain number of values, or when we are counting something (using whole numbers). For

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example, 5 students, 10 trees etc.

Question: What are digital circuits?

Answer: Digital circuits are electric circuits based on a number of discrete voltage levels.

In most cases there are two voltage levels: one near to zero volts and one at a higher level depending on the supply voltage in use. These two levels are often represented as L and H.

Question: What is absurdity or contradiction?

Answer: A statement which is always false is called an absurdity.

Question: What is contingency?

Answer: A statement which can be true or false depending upon the truth values of the variables is called a contingency.

Question: Is there any particular rule to solve Inductive Step in the mathematical Induction?

Answer: In the Inductive Step, we suppose that the result is also true for other integral values k. If the result is true for n = k, then it must be true for other integer value k +1 otherwise the statement cannot be true.
In proving the result for n = k +1, the procedure changes, as it depends on the shape of the given statement.
Following steps are main:

1) You should simply replace n by k+1 in the left side of the statement.
2) Use the supposition of n = k in it.
3) Then you have to simplify it to get right side of the statement. This is the step,
where students usually feel difficulty.
Here sometimes, you have to open the brackets, or add or subtract some terms
or take some term common etc. This step of simplification to get right side of the given statement for n = n + 1 changes from question to question.
Now check this step in the examples of the Lessons 23 and 24.

Question: Different notations of conditional implication.

Answer: If p than q. P implies q. If p , q. P only if q. P is sufficient for q.

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Question: What is cartesion product?

Answer: Cartesian product of sets:- Let A and B be sets. The Cartesian product of A and B, denoted A x B (read "A cross B") is the set of all ordered pairs (a, b), where a is in A and b is in B. For example: A = {1, 2, 3, 4, 5, 6} B = {a} A x B = {(1,a), (2,a), (3,a), (4,a), (5,a)}

Question: Define fraction and decimal expansion.

Answer: Fraction:- A number expressed in the form a/b where a is called the numerator and b is called the denominator. Decimal expansion:- The decimal expansion of a number is its representation in base 10 The number 3.22 3 is its integer part and 22 is its decimal part The number on the left of decimal point is integer part of the number and the number on the right of the decimal point is decimal part of the number.

Question: Explane venn diagram.

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common.

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Question: Write the types of functions.

Answer: Types of function:- Following are the types of function 1. One to one function 2. Onto function 3. Into function 4. Bijective function (one to one and onto function) One to one function:- A function f : A to B is said to be one to one if there is no repetition in the second element of any two ordered pairs. Onto function:- A function f : A to B is said to be onto if Range of f is equal to set B (co-domain). Into function:- A function f : A to B is said to be into function of Range of f is the subset of set B (co domain) Bijective function: Bijective function:- A function is said to be Bijective if it is both one to one and onto.

Question: What is conditional probability with example?.

Answer: Click here.

Question: Explain combinatorics.

Answer: Branch of mathematics concerned with the selection, arrangement, and

combination of objects chosen from a finite set.

The number of possible bridge hands is a simple example; more complex problems include scheduling classes in classrooms at a large university and designing a routing system for telephone signals. No standard algebraic procedures apply to all combinatorial problems; a separate logical analysis may be required for each problem.

Question: How the tree diagram use in our real computer life?

Answer: Tree diagrams are used in data structure, compiler construction, in making algorithms, operating system etc.

Question: Write detail of cards.

Answer: Diamond Club Heart Spade A A A A 1 1 1 1 2 2 2 2 3 3 3 3 4 4 4 4 5 5 5 5 6 6 6 6 7 7 7 7 8 8 8 8 9 9 9 9 10 10 10 10 J J J J Q Q Q Q K K K K Where 26 cards are black & 26 are red. Also 'A' stands for 'ace' 'J' stands for 'jack' 'Q' stands for 'queen' 'K' stands for 'king'

Question: what is the purpose of permutations?

Answer: Definition:- Possible arrangements of a set of objects in which the order of the arrangement makes a difference. For example, determining all the different ways five books can be arranged in order on a shelf. In mathematics, especially in abstract algebra and related areas, a permutation is a bijection, from a finite set X onto itself. Purpose of permutation is to establish significance without assumptions

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Question: How a compound statement is formed.

Answer: A compound statement is made from two or more statements through some logic operator. for example "I am happy",," she is angry" are two statements. We can join these two to form a statement which will be compound statement. If we use a conjunction (and) to join these two statements, the compound statement will be "I am happy and she is angry". if p= I am happy q=She is angry the compound statement will be p^q = "I am happy and she is angry"

Question: Explain the inclusion-exclusion principle.

Answer: Inclusion-Exclusion principle for two sets: The Principle itself can also be expressed in a concise form. It consists of two parts. The first just states that counting makes sense. The second describes its fundamental property. Every group of objects A can be associated with a quantity - denoted n (A) - called the number of elements in A. If A and B has disjoint finite sets, then n (A union B) = n (A) + n (B) (by disjoint we means(A intersection B) = empty set) The above is the condition when A and B are disjoint but the next

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question which arises in our mind if A and B are not disjoint sets then what happen? If A and B are disjoint sets , we get the simplest form of the Inclusion-Exclusion Principle as n (A union B) = n (A) + n (B) - n (a intersection B) Indeed , in n (A) + n (B) some elements counted twice. which are they? The elements that were counted twice are exactly those that belong to A (one count) and also belong to B (the second count). In short, counted twice were the elements of A intersection B . To obtain an accurate number n (A union B) of elements in the union we have to subtract from n (A) + n (B) the number n (A intersection B) of such elements. Inclusion-Exclusion Principle for three sets : If A , B ,C are finite disjoint sets then n (A union B union C) = n (A) + n (B) + n (C) (by disjoint we means(A intersection B intersection C) = empty set) If A , B ,C are finite sets then n (A union B union C) = n (A) + n (B) + n (C) - n (A intersection B) - n (B intersection C) - n (A intersection C) + n (A intersection B intersection C) Lets see the example Example Counting the number of elements in an Intersection. In a class of students undergoing a computer course the following were observed. Out of a total of 50 students: 30 know Pascal, 18 know Fortran, 26 know COBOL, 9 know both Pascal and Fortran, 16 know both Pascal and COBOL, 8 know both Fortran and COBOL, 47 know at least one of the three languages. From this we have to determine a. How many students know none of these languages? b. How many students know all three languages? a. We know that 47 students know at least one of the three languages in the class of 50. The number of students who do not know any of three languages is given by the difference between the number of students in class and the number of students who know at least one language. Hence 50 - 47 = 3 b. Let us assume that A = All the students who know Pascal in class. B = All the students who know COBOL in the class. C = All the students who know FORTRAN in class. n (A) = 30 n (B) = 26 n (C) = 18 n (A intersection B) = 16 n (B intersection C) = 18 n (A intersection C) = 9 By applying the Inclusion exclusion principle, n (A union B union C) = n (A) + n (B) + n (C) - n (A intersection B) - n (B intersection C) - n (A intersection C) + n (A intersection B intersection C) By substituting the values 47 = 30 + 26 + 18 -16 -18 -9 + n (A intersection B intersection C) Solving for n (A intersection B intersection C) , we get. n (A intersection B intersection C) = 6 Hence there are 6 students in the class who know all the three languages.

Question: What is generalized pigeonhole and pigeonhole principle?

Answer: Let us take a function f : A---B where A be the domain set and B be the co-domain set. If n(A) <= n(B),then f is called the one-one function. But in pigeonhole principle, there is a rule/function from the finite domain set to the smaller co-domain set, and it should not be one-one (i.e. n(A)>n(B)). That is there must be atleast two element in the domain set A which have the same images in the co-domain set B. (i.e. at least 2(say a and b) elements in set A have the images (let say 1) in set B). Now we discuss a very simple

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example, given in handouts. Given any set of seven integers, must there be two that have the same remainder when divided by 6? Solution: Let we take a domain set A=a set of any seven integers= {6, 7, 8, 9, 10, 11, 12, 13} And the co-domain set say B=a set of the remainder when integer divided by 6= {0, 1, 2, 3, 4, 5, 6} Then the set of possible remainders that can be obtained when the integers Like {6, 7, 8, 9, 10, 11, 12, 13} ,divided by 6 contains the remainder set {0, 1, 2, 3, 4, 5}. This set has 6 elements. Thus by the pigeonhole principle if 7 = 6 + 1 integers, each are divided by six, then at least two of them must have the same remainder.(As take 6, 7, 8, 9, 10, 11, 12, 13) have atleast two integers which have the same remainder(i.e. 7 and 13 have the same remainder 1). In Generalized pigeonhole principal, It states that for n elements from the domain sets(or n pigeons),there is m elements in the co domain set(i.e. m pigeonholes),and for some +ve integer k, n>k.m, then there is at least one pigeonhole which contains k+1 or more pigeons. Note: As in pigeonhole principle, atleast one pigeonhole must contain two or more pigeons while in generalized atleast one pigeonhole must contain k+1 or more pigeons]. Or we say that If for n pigeons fly into the m pigeonholes and, for some +ve integer k, n>k.m, then there is at least one pigeonhole which contains k+1 or more pigeons.

Question: What are Binary Search trees?

Answer: In programming, Binary Search tree means "a specific type of tree data structure in which each node has at most two sub trees, one left and one right." Binary trees are often used for sorting information; each node of the binary search tree contains a key, with values less than that key added to one sub tree and values greater than that key added to the other.

Question: What are the basic steps of probability?

Answer: The basic steps of probability are as under; 1. First list out all possible out comes. That is called the sample space S For example when we roll a die the all possible outcomes are the set S i.e. S = {1,2,3,4,5,6} 2. Secondly we have to find out all that possible outcomes, in which the probability is required. For example we are asked to find the probability of even numbers. First we decide any name of that event i.e E Now we check all the even numbers in S which are E = {2,4,6} Remember Event is always a sub-set of Sample space S. 3. Now we apply the definition of probability P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) So from above two steps we have n (E) = 3 and n (S) = 6 then P(E) = 3 / 6 = 1/2 which is probability of an even number.

Question: What is the main purpose of probability?

Answer: The purpose of probability theory is to give a precise mathematical meaning to randomness. Doing this requires a certain amount of abstraction. It is always clear what we mean when make a statement of the type that it is very

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likely to rain or I have a fair chance of passing the annual examination Or A(i.e. a event) will probably win a prize , etc. In each statement uncertainty is expressed , but on the basis of past evidence , we have some degree of personal belief in the truth of each statement Today the probability theory has a wider field of application and is used to make intelligent decision in economics, Management, operations Research, Sociology, Psychology, Astronomy , Physics , Engineering and Genetics where risk and uncertainty are involved . Probability these days is best understood through the application of modern set theory.

Question: If two events are disjoint then what will be the probability formula for union?

Answer: If A and B are two disjoint (mutually exclusive) events of a sample space S, then P (AUB) = P(A) + P(B)

Question: If two events are not disjoint then what will be the probability formula for union.

Answer: If A and B are any two events (which are joint) of a sample space S, then P(AUB) = P(A) + P(B) - P(A^B)

Question: What will be the maximum probability of an event?

Answer: The maximum probability of an event may be 1

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Question: What will be the minimum probability of an event?

Answer: The minimum probability of an event may be 0

Question: What are dependent and independent events?

Answer: Independent events:- Two events are said to be independent if the result of the second event is not affected by the result of the first event. If A and B are independent events, the probability of both events occurring is the product of the probabilities of the individual events. If A and B are independent events, P(A and B) = P(A) x P(B). Example: A drawer contains 3 red paperclips, 4 green paperclips, and 5 blue paperclips. One paperclip is taken from the drawer and then replaced. Another paperclip is taken from the drawer. What is the probability that the first paperclip is red and the second paperclip is blue? Because the first paper clip is replaced, the sample space of 12 paperclips does not change from the first event to the second event. The events are independent. P(red then blue) = P(red) x P(blue) = 3/12 • 5/12 = 15/144 = 5/48. Dependent events:- If the result of one event is affected by the result of another event, the events are said to be dependent. If A and B are dependent events, the probability of both events occurring is the product of the probability of the first event and the probability of the second event

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once the first event has occurred. If A and B are dependent events, and A occurs first, P(A and B) = P(A) x P(B,once A has occurred) Example: A drawer contains 3 red paperclips, 4 green paperclips, and 5 blue paperclips. One paperclip is taken from the drawer and is NOT replaced. Another paperclip is taken from the drawer. What is the probability that the first paperclip is red and the second paperclip is blue? Because the first paper clip is NOT replaced, the sample space of the second event is changed. The sample space of the first event is 12 paperclips, but the sample space of the second event is now 11 paperclips. The events are dependent. P(red then blue) = P(red) x P(blue) = 3/12 • 5/11 = 15/132 = 5/44

Question: Define multiplication and give example.

Answer: Multiplication theorem: Let A and B be any two events defined in a sample space S, then P (A intersection B) = P (A) P(B/A) ,Provided P(A) not equal to zero Or P (A intersection B) = P (B) P(A/B) ,Provided P(B) not equal to zero The conditional probability of B given that A has occurred is P (B/A) = P (A intersection B)/P (A) ,Provided that P (A) not equal to zero. Multiplying both sides by P (A), we get P(A intersection B) = P (A) * P (B/A) The second form is easily obtained by interchanging A and B. This is called general rule of Multiplication of Probabilities. For example : Two cards are dealt from a deck from a pack of ordinary cards. Find the probability that the second card dealt is a heart? SOLUTION: Let H1 is the event that the first card dealt is a heart and H2 is the event that the second event dealt is a heart. P (H1) = 13/52 P(not H1) = 1- P(H1) = 1- 13/52 = 39/52 Then P (Second card is a heart) = P (First card is a heart and second card is a heart) + P (first card is not a heart and second card is a heart) = P (H1 intersection H2) + P(not H1 intersection H2) = P (H1) P(H2/H1) + P (Not H1) P(H2/not H1) = 13/52 * 12/51 + 39/52*13/51 = 1/17 + 13/68 = 17/68

Question: When will we use conditional probability?

Answer: In Conditional probability we put some condition on an event to be occur. e.g. A pair of dice is tossed. Find the probability that one of the dice is 2 if the sum is 6. If we have to find the probability that one of the dice is 2, then it is the case of simple probability. Here we put a condition that sum is six. Now A = { 2 appears in atleast one die} E = {sum is 6 } Here E = { (1,5), (2, 4), (3, 3), (4, 2), (5, 1) } Here two order pairs (2, 4) and (4, 2) satisfies the A. (i.e. belongs to A) Now A (intersection) B= { (2,4), (4,2)} Now by formula P(A/E) = P(A (intersection) E)/ P(E) = 2/5

Question: What is conditional probability?

Answer: Conditional Probability The probability of an event occurring given that another event has already occurred is called a conditional probability. The probability that event B occurs, given that event A has already occurred is

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P(B|A) = P(A and B) / P(A) Similarly the probability that event A occurs, given that event B has already occurred is P(A|B) = P(A and B) / P(B)

Question: Explain the intersection between probability sets?

Answer: We can multiply probabilities of dependent as well as independent events. Two events "A" and "B " are said to be independent , if the occurrence of any one of then does not influence the occurrence of the other event. Now if events are independent then P(A^B) = P(A).P(B) Example: if P(A) = 0.8 and P(B) = 0.75 here remembering that both A and B are independent then P(A^B) = 0.8 x 0.75 = 0.6

Question: What is random variable?

Answer: Random variable : A random variable is a rule that assigns numerical value to each outcome in a sample space S. Or A numerical quantity, whose value is determined by the outcome of a random experiment, is called the random variable". A random variable is also called a chance variable.

Question: What is Probability distribution of a random variable?

Answer: Probability distribution of a random variable: Let X be a discrete random variable which assume values x1 ,x2,x3,...xn,...Then the function ,denoted by p (x) or f (x) , and defined by f (xi) = P (X = xi) for all i = 1,2,3,...n,... is called the Probability function or Probability distribution of random variable X and the values x1 ,x2,x3,...xn,...among which the total probability 1 is distributed, are called the probability points , P (X =xi) means the probability that the discrete random variable X takes values xi. A probability distribution must satisfy the following two properties 1. f (xi) greater than or equal to zero for all i 2. Sum of probabilities f (xi) is equal to 1 For example: A random variable assume values 0 ,1, 2 , 3 when a fair coin is tossed three times and their probabilities are f (0) = P (X = 0) = P [{TTT}] = 1/8 f (1) = P (X = 1) = P [{HTT , THT, TTH}] = 3/8 f (2) = P (X = 2) = P [{HHT , HTH, THH}] = 3/8 f(1) = P (X = 3) = P [{HHH}] = 1/8 Putting the information in the tabular form, we obtained the desired probability distribution of X.

Question: What is expected value of random variable?

Answer: Expected Value The expected value (or population mean) of a random variable indicates its average or central value. It is a useful summary value (a number) of the variable's distribution. Stating the expected value gives a general impression of the behavior of some random variable without giving full details of its probability distribution (if it is discrete) or its probability density function (if it is continuous). Two random variables with the same expected value can have very different distributions. There are other useful descriptive measures which affect the shape of the distribution, for example

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variance. The expected value of a random variable X is symbolized by E(X) or μ. If X is a discrete random variable with possible values x1, x2, x3, ..., xn, and p(xi) denotes P(X = xi), then the expected value of X is defined by: μ=E(X)=?xi p(xi) where the elements are summed over all values of the random variable X. If X is a continuous random variable with probability density function f(x), then the expected value of X is defined by: μ=E(X)= ?x f(x)dx Example Discrete case : When a die is thrown, each of the possible faces 1, 2, 3, 4, 5, 6 (the xi's) has a probability of 1/6 (the p(xi)'s) of showing. The expected value of the face showing is therefore: μ = E(X) = (1 x 1/6) + (2 x 1/6) + (3 x 1/6) + (4 x 1/6) + (5 x 1/6) + (6 x 1/6) = 3.5 Notice that, in this case, E(X) is 3.5, which is not a possible value of X.

Question: what are vertices, edges, incident, adacent, loop, parallel and isolated?

Answer: Vertices are nodes or points and edges are lines/arcs which are used to connect the vertices. e.g If you are making the graph to find the shortest path between the cities Lahore, Islamabad, Faisalabad , Karachi, and Multan. Then cities Lahore, Islamabad, Faisalabad, Karachi, and Multan are vertices and roads between them are edges. Two vertices that are connected by an edge are called adjacent vertices. An edge is said to be incident on each of its endpoints. A loop is an edge that connects a vertex to itself. A vertex on which no edges are incident is called isolated. Two distinct edges with the same set of end points are said to be parallel. Q#19 what is degree of vertex? Degree Of A Vertex:- In graph theory, the degree of a vertex of a graph is the number of edges incident to the vertex, with loops counted twice. The degree of a vertex v is denoted deg(v).

Question: What is bipartite graph?

Answer: A bipartite graph G is a simple graph (i.e it does not have any loop or parallel edges) whose vertex set can be partitioned into two mutually disjoint non empty subsets A and B such that the vertices in A may be connected to vertices in B, but no vertices in A are connected to vertices in A and no vertices in B are connected to vertices in B. i.e. The vertices of this graph can be divided into two disjoint subsets. And vertices in one set cannot be connected to each other but with the vertices of other sets.

Question: It is compulsory in Euler circuits that every vertex has even degree?

Answer: Yes, it is compulsory in Euler circuits that every vertex has even degree.

Question: What is Hamiltonian circuit?

Answer: A tour of the vertices of a graph which begins and ends at the same vertex of the graph and which visits each vertex of the graph once and only once

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moving along distinct edges of the graph. In the mathematical field of graph theory, a Hamiltonian path is a path in an undirected graph which visits each vertex exactly once. Examples 1) A complete graph with more than two vertices is Hamiltonian 2) Every cycle graph is Hamiltonian

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Question: Define isomorphic invariant.

Answer: Let G and G' be two graphs such that G' is isomorphic to G. Let P be a property that G and G' must both have then such a property P is said to be an isomorphic invariant.

Question: Is there are two loops in any vertex?

Answer: The vertex has only one loop and the degree of loop is 2.

Question: What is a method of finding a circuit of a graph?

Answer: Circular: A circular path in which electricity travels is called circular. Open Circuit: A circuit having some breaks in it so that current is not passing or cannot pass. An Euler circuit for G is a circuit that contains every vertex and every edge of G. It is a sequence of adjacent vertices and edges in G that starts and ends at the same vertex, uses every vertex of G at least once, and uses every edge of G exactly once. A Hamiltonian path is a path that contains every vertex of G. A Hamiltonian circuit is a simple circuit that includes every vertex of G. Other properties: If graph G has a Hamiltonian circuit, then G has a sub graph H with the following properties: 1) H contains every vertex of G; 2) H is connected. 3) H has the same number of edges as vertices. 4) Every vertex of H has degree 2. You can use appropriate method according to the mention method or condition.

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Question: What is the difference between edge and vertex?

Answer: Vertices are nodes or points and edges are lines/arcs which are used to connect the vertices. e.g If you are making the graph to find the shortest path or for nay purpose of cites and roads between them which contain Lahore, Islamabad, Faisalabad , Karachi, and Multan. Then cities Lahore, Islamabad, Faisalabad, Karachi, and Multan are vertices and roads between them are edges.

Question: What is the difference between yes and allowed in graphs?

Answer: Allowed mean that specific property can be occurs in that case but yes mean that specific property always occurs in that case. e.g. In Walk you may start and end at same point and may not be (allowed). But in Closed Walk you have to start and end at same point (yes).

Question: What is simple circuit?

Answer: A path of length n from u to v , where n is a positive integer, in an undirected graph is a sequence of edges e1,e2,e3,.....en of the graph such that f(e1) = {x0,x1},f(e2) = {x1,x2},f(e3) = {x2,x3},.....,f(en) = {xn-1,xn}, where x0 = v . When the graph is simple, we denote this path by its vertex sequence x0,x1,x2,.....xn, (Since listing these vertices uniquely determines the path). The path is a circuit if it begins and ends at the same vertex , that is , if u = v . The path or circuit is said to pass through or traverse the vertices x1,x2,x3,.....xn-1, A path or circuit is simple if it does not contain the same edge more than once.

Question: What is difference between walk and circuit?

Answer: WALK, A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G. Thus a walk has the form v0e1v1e2 … vn - 1envn Where the v's represent vertices, the e's represent edges v0=v , vn=w, and for all i = 1, 2 … n, vi-1 and vi are endpoints of ei. The trivial walk from v to v consists of the single vertex v. CIRCUIT, A circuit is a closed walk that does not contain a repeated edge. Thus a circuit is a walk of the form v0e1v1e2 … vn-1 en vn Where v0 = vn and all the ei,s are distinct.

Question: What is the four color theorem and what is the use of color graphs?

Answer: Graph coloring is an interesting application to look at for several reasons. First, it is in the class of problems known as NP-Complete, even for planar graphs. In addition, a large number of practical problems can be formulated in terms of coloring a graph, including many scheduling problems. Quite a bit of research has been extended in this area, including the well-known theorem that every planar graph can be 4-colored. Finally, it is quite easy to make up a large range of problems, from simple to complex, single dimensional to multi-dimensional. Or A k coloring of a graph is legal if two adjacent vertices are not assigned the same color The minimum k colors used are called the chromatic number. The graph coloring problem is to find the minimum k for a legal coloring to occur. The uses of graph coloring are Scheduling 1. Class exam assignments for courses 2. Let the vertices be a set of courses and the edges be a pair of courses whose exams must not be at the same time Frequency assignment 1. let the frequencies be assigned as the vertices and the cell phones assigned as the edges 2. For this we want to minimize the number of frequencies used for the cell phones Register Allocation 1. Variables are assigned to a number of hardware registers 2. Multiple variables can be assigned to registers since there are far more variables than registers 3. Conflicting variables occur when variables are used right away before or after another 4. Assign variables that do not conflict with one another to minimize the use of non-register memory 5. For graph coloring, assign the variables as nodes and edges as conflicts between nodes 6. A coloring for this assignment is conflict free Note: A vertex is colored one color and any vertex adjacent is colored a different color.

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Question: What is the principal of Mathematical Induction?

Answer: Mathematical induction is one of the most recently developed techniques for a proof .It is most frequently used in computer science. Let P(n) be a predicate defined for all positive integers n. P(n) is true for all positive integer n greater then or equal to 1 if the following two steps are satisfied 1.Basic Step P(1) is true This means that we put n=1 in the propositional function and check whether the left hand side is equal to right hand side or not. If both sides are equal then Basic step is satisfied 2.Inductive Step If P(k) is true then P(k+1) is true for all k greater then or equal to1. This means that we suppose that the propositional function is true for n=k then we have to prove that it is true for n=k+1.If we are able to prove it then the inductive step is satisfied. For example we have to find the sum of first n positive integers. We have the proposition function P(n) as 1+2+3+...+n=n(n+1)/2----------(1) Basic step is clear to you i only explain inductive step. In inductive step we first suppose that P(n) is true for n=k we get 1+2+3+...+k=k(k+1)/2---------(2) Now we have to prove that P(n) is true for n=k+1.Replacing n by k+1 in (1) we have 1+2+3+...k+1=(k+1)((k+1)+1)/2 Re write the left hand side of the above equation 1+2+3+...k+k+1 (because the proceeding term of k+1 is k) 1+2+3+...+k+(k+1)=k(k+1)/2+(k+1) By using (2) and simplifying you get the required result.

Question: What is Identity function?

Answer: An identity function, also called identity map or identity transformation, is a function that always returns the same value that was used as its argument. In terms of equations, the function is given by f(x)= x.

Question: What is a Direct Proof?

Answer: A direct proof is a way of showing the truth or falsehood of a given statement by a straightforward combination of established facts, usually existing lemmas and theorems, without making any further assumptions. In order to directly prove a conditional statement of the form "If p, then q", it is only necessary to consider situations where the statement p is true. Logical deduction is employed to reason from assumptions to conclusion.

Question: What are the basic methods for indirect proof?

Answer: There are two methods: 1: Proof by Contraposition. 2: Proof by Contradiction.

Question: How to prove by contraposition?

Answer: If we have a statement of the form A implies B or A ? B (If A then B), then the contra positive is: not B implies not A or ~ B ? ~ A. A statement is always equivalent to its contra positive. If you have proved one, you have

proved the other.

Question: How to proof by Contradiction?

Answer: To prove something by contradiction, we assume that what we want to prove is not true, and then show that the consequences of this are not possible. That is, the consequences contradict either what we have just assumed, or something we already know to be true (or, indeed, both). We call this a contradiction.

Question: what is the meaning of declarative sentence?

Answer: Declarative sentences are the sentences that state some fact. Sentences that state a fact are: Islamabad is the capital of Pakistan 2+2=4 He is very rich Sentences that don't state a fact are: What time is it? Read this book carefully Close the door

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Question: What is Constant function?

Answer: A constant function is a function whose values do not vary and thus are constant. Example: If f(x) = 2 for any x, then f is constant since any value in the domain is mapped to 2.

Question: What is Loop Invariant?

Answer: The loop invariant is a method which is used to prove the correctness of the loop with respect to certain per and post conditions. It is based on the principle of mathematical induction.

Question: Difference between contradiction and contingency.

Answer: A contradiction is a statement form that is always false regardless of the truth values of the statement variables whereas a statement which can be true or false depending upon the truth values of the variables is called a contingency.

Question: What is K-Sample?

Answer: A k sample of a set of n elements is a choice of k elements taken from the set of n elements such that the order matters and the elements can be repeated. In k sample the repetition of elements are allowed. Suppose there are n distinct elements and we draw a k sample from it. The first element of k sample can be drawn in n ways. Since the repetition is allowed so the second can also be drawn in n ways. The third element can be drawn in n ways. Similarly the k element can be drawn in n ways. Since by Product Rule, the total number of ways in which k sample can be selected from n distinct elements is n.n.n...n (k times) = n k

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Question: What is Hierarchy of operations for logical connectives?

Answer: Hierarchy determines which Logical Connectives evaluated first. To compute the truth values for a statement form, follow rules similar to those used to evaluate algebraic expressions. For each combination of truth values for the statement variables, first evaluate the expressions with in the inner most parentheses, then evaluate the expression with in the next innermost set of parentheses, and so forth until you have the truth values for the complete expression Following is the order of evaluation • ~ (negation) • ^(conjunction), v(disjunction) • --> (conditional), , <-->(biconditional) The statements (p ^ q) v r is not equivalent to p ^ (q v r)

Question: What is Excusive OR?

Answer: Excusive OR also called Exclusive disjunction is a type (symbolized as XOR) of logical operator that results in a value of "true" if and only if exactly one of the operands has a value of "true". The Truth table for p XOR q p q p XOR q T T F T F T F T T F F F For example, in a statement like "You may have cold drink or you may have tea" tells that exactly one of the conditions can be true at a time.you can't have both cold drink and tea at the same time. So, here , XOR will be used If we have two statements p= "Ali's height is 6 feet " q="24 is divisible by 4 " Now , here , both statements can be true at the same time so OR will be used.

Question: what is plus sign within circle is called and what this symbol is called?

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Answer: This is a logical operator and is a symbol of exclusive disjunction, also called exclusive or (symbolized XOR or EOR), is a type of logical disjunction.

Question: Alternative ways of expressing biconditional statement.

Answer: •P is necessary and sufficient for q •If p then q, and conversely •P is equivalent to q

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Question: Difference between argument and premise.

Answer: when some statement is said on the basis of a set of other statements, meaning that this statement is derived from that set of statements, this is called an argument. The formal definition is "an argument is a list of statements called premises (or assumptions or hypotheses) which is followed by a statement called the conclusion. " All statements except the last statement is called Premises (or hypothesis or assumptions) So argument is made of premises and conclusion. The last and the final statement is called the conclusion. For example We have an argument as If it rains in the evening then we will go for jogging. If we will go for jogging then we will be tired. ----------

Therefore if it rains then we will be tired. These two statements “If it rains in the evening then we will go for jogging.” “If we will go for jogging then we will be tired.” are premises of the given argument. And the statement “Therefore if it rains then we will be tired” is the conclusion of the given argument.

Question: What is critical row for finding valid or invalid argument?

Answer: Critical rows are those in which the premises are true, now if conclusion is also true then it means the statement is valid but if you find only one such row in which all premises are true but conclusion is false then statement is invalid. That's why we have to see the critical row.

Question: What is boolean expression?

Answer: A Boolean expression is an expression that results in a Boolean value, that is, TRUE or FALSE. For example, the value for 5 > 3 is TRUE, the value for "An apple is not a fruit" is FALSE.

Question: What is universal set?

Answer: The set that contains all the elements or objects involved in the problem under consideration is called the universal set. For Example, 1) In plane geometry, the universal set consists of all the points in the plane, 2) In human population studies the universal set consists of all the people in the world. We will let the symbol U denote the universal set. 3) Let U= {1, 2, x, 3, 4, y, 6, 7, z, 9} A== {1, 2, x, 3, 4} B= {3, 4, y, 6, 7, z, 9} Here U is the universal set because the set U is the superset related to sets A and B i.e. The set A and B are the subsets of set A and B.

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Question: What is membership table and truth table?

Answer: Membership table: A table displaying the membership of elements in sets. Set identities can also be proved using membership tables. An element is in a set, a 1 is used and an element is not in a set, a 0 is used. Truth table: A table displaying the truth values of propositions.

Question: What is partion in terms of sets?

Answer: A partition of a set X is a set of nonempty subsets of X such that every element x in X is in exactly one of these subsets. Equivalently, a set P of subsets of X, is a partition of X if The union of the elements of P is equal to X. (We say the elements of P cover X.) The intersection of any two elements of P is empty. (We say the elements of P are pair wise disjoint.) The elements of P are sometimes called the blocks or parts of the partition. Examples Every singleton set {x} has exactly one partition, namely { {x} }. For any nonempty set X, P = {X} is a partition of X. For any non-empty proper subset A of a set U, this A together with its complement is a partition

of U. The set { 1, 2, 3 } has these five partitions. { {1}, {2}, {3} }, sometimes denoted by 1/2/3. { {1, 2}, {3} }, sometimes denoted by 12/3. { {1, 3}, {2} }, sometimes denoted by 13/2. { {1}, {2, 3} }, sometimes denoted by 1/23. { {1, 2, 3} }, sometimes denoted by 123. Note that { {}, {1,3}, {2} } is not a partition (because it contains the empty set). { {1,2}, {2, 3} } is not a partition (of any set) because the element 2 is contained in more than one distinct subset. { {1}, {2} } is not a partition of {1, 2, 3} because none of its blocks contains 3; however, it is a partition of {1, 2}.

Question: What does the cardinality of a set?

Answer: In mathematics, the cardinality of a set is a measure of the "number of elements of the set". For example, the set A = {1, 2, 3} contains 3 elements, and therefore A has a cardinality of 3. The cardinality of a set A is denoted |A|, with a vertical bar on each side; this is the same notation as absolute value and the meaning depends on context.

Question: What are invertible functions or inverse functions?

Answer: Since inverse of every function does not exist. It exists only for bijective functions. An inverse or invertible function must satisfy the condition that all of the elements in the domain and range are paired-up in monogamous relationship that is each element in the domain pairs to only one element in the range and each element in the range pairs to only one element in the domain.

Suppose $f:X \rightarrow Y$ is a bijective function. Then the inverse function $f^{-1}: Y \rightarrow X$ is defined as:

$\forall\ y \in Y, f^{-1}(y) = x \Leftrightarrow y = f(x)$.

Question: When a function is Bijective (One-to-one correspondence)?

Answer: A bijective function is a function f from a set X to a set Y with the property that, for every y in Y, there is exactly one x in X such that $f(x) = y$ and no unmapped element remains in both X and Y.

OR

A function which is both one-to-one (injective) and onto (surjective) is called a bijective function or a one-to-one correspondence.

Example: The function f from the real line **R** to **R** defined by $f(x) = 2x + 1$ is one-one and on-to.

One-one: Let $f(x_1) = f(x_2)$ for $x_1, x_2 \in R$

$\Rightarrow$ $2x_1 + 1 = 2x_2 + 1$ (by definition of f)

$\Rightarrow$ $2x_1 = 2x_2$ (subtracting 1 on both sides)

$\Rightarrow$ $x_1 = x_2$ (dividing by 2 on both sides)

Hence f is one-to-one.

On-to: Since for each y there is a unique $x = (y - 1)/2$ such that $f(x) = y$.

So f is on-to.

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Thus f is bijective.

Question: What is the composition of functions?

Answer: let f: X→Y′ and g:Y→Z be two functions with condition that the range set of f is the subset of the domain set of g. To find the composition we use the notation (f o g) (x). Another way to write this is f(g(x)).
Example: Find both: (a) (f o g)(x) and (b) (g o f)(x).

Where f(x) = 4x-2 and g(x) = x^2 + 2.
(g o f) (x) = g (f(x)) = g (4x-2) = (4x-2)^2 +2.
= (16 x^2 – 16x + 4) + 2 = 16 x^2 -16x +6.
(f o g)(x) = f (g(x)) = f (x^2 +2) = 4(x^2 +2) -2.
= (4x^2 +8) -2 = 4x^2 +6.

Question: What is meant by factorial?

Answer: The factorial of a non-negative integer n, denoted by $n!$, is the product of all positive integers less than or equal to n. For example,

$$6! = 6*5*4*3*2*1 = 270$$

And

$$4! = 4*3*2*1 = 24$$

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Question: What is k-permutation?

Answer: A k permutation of a set of n elements is a choice of k elements taken from the set of n elements such that the order matters and the elements can not be repeated.
In k permutation the repetition of elements are not allowed. Suppose there are n distinct elements and we draw a k permutation from it. The first element of k permutation can be drawn in n ways. Since the repetition is not allowed so the second can be drawn in n-1 ways. The third element can be drawn in n-2 ways. Similarly the kth element can be drawn in (n-(k-1)) ways. Since by Product Rule, the total number of ways in which k permutation can be selected from n distinct elements is

P(n,k) = n . (n - 1) . (n - 2) … (n - (k-1))
= n . (n - 1) . (n - 2) … (n – k + 1)
= [n . (n-1) . (n -2) … (n – k +1)][(n – k) (n – k -1) … 3 . 2 .1]
[(n – k) (n – k -1) … 3 . 2 . 1]
= n! / (n –k)!

Question: What are the different forms of a set?

Answer: Click here to view the different forms of a set.

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Question: Explain contrapositive.

Answer: The contrapositive of the conditional statement p --> q is ~ q --> ~ p i.e. Take the negation of conclusion and made it hypothesis and similarly take the negation of conclusion and made it conclusion. Where p and q are propositions, in other words, contrapositive reverses the roles of p and q and negates each of them e.g. 1. If the cat will run then the dog will chase the cat.....conditional If the dog will NOT chase the cat then the cat will NOT run ...contra positive 2. If p then q ....conditional If negation q then negation p .....contra positive

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